
PROJECT PROPOSAL

1. Title of the Project

Predictive Maintenance of Industrial Equipment Using Machine Learning

2. Brief on the Project

- **Project Type:** Predictive Data Analytics and Supervised Machine Learning.
- **Problem Addressed:** Industrial manufacturing faces massive financial losses due to unplanned equipment downtime and sudden machine failures.
- **Motivation:** Traditional schedule-based maintenance either replaces working parts prematurely or fails to catch sudden breakdowns. Using data-driven insights minimizes operational costs and prevents catastrophic gear failure.
- **Previous Work:** Industry standard practices rely heavily on basic thresholds for sensor signals (e.g., stopping a machine if temperature reaches 90°C). Modern studies lean toward training multi-sensor algorithms (Random Forests) to predict Remaining Useful Life (RUL) and precise failure modes.
- **Tentative Approach:** We will construct a binary and multi-class classification engine using historical telemetry sensor logs. The framework will ingest data, filter noise, construct rolling temporal statistics, and use tree-based ensembles to predict machine statuses.

3. Deliverables of the Project

We will evaluate classification success using a decoupled data test split. Model validation metrics will rely on **Recall** (to minimize false negatives/unpredicted breakdowns) and **F1-Score** (to balance false alarms).

List of Questions Your Model/Problem Are Designed to Answer:

- Is this specific machine highly likely to fail within the next 24 operational hours?
- Which exact component (e.g., tool wear, heat dissipation, power overload) is causing the critical risk?
- Which sensor telemetry features correlate most strongly with imminent structural failure?

- Provided data Consists of Class imbalance issue

Details of the Model, Important Findings, and Expected Outcomes:

- **Model Choices:** Random Forest Classifier and Extreme Gradient Boosting (XGBoost) for structural sensor logs.
 - **Expected Findings:** High structural degradation corresponds heavily with intersecting spikes in operational temperature and rotational speed.
 - **Expected Outcome:** A predictive system that accurately flags 90%+ of mechanical faults ahead of time, dropping baseline corporate maintenance expenditure.
-

4. Resources

- **Data Set Source:** Download data from Advance certification Portal
- **Software:**
 - Python 3.x ecosystem
 - Pandas & NumPy (Data parsing)
 - Scikit-Learn & XGBoost (Machine learning architecture)
 - Matplotlib & Seaborn (Data visualization) [6, 7]

- **References:**

1. *Predictive Maintenance of Machine Tool Systems Using Artificial Intelligence.* ScienceDirect.
 2. *A Machine Learning Implementation to Predictive Maintenance and Monitoring of Industrial Compressors.*
 3. *Predictive Maintenance in Industrial Systems Using Machine Learning*
 4. *Learn from data and concept of predictive maintenance on internet*
-

5. Individual Details

- **Name:** [Nitin Baban Devekar]
 - **E-mail ID:** [nitindevekar@gmail.com]
 - **Phone Number:** [+91 8805182969]
-

6. Milestones

Milestone Track	Activity Description	Expected Timeline
Define a problem	Formalize targets for equipment failures and business scope.	27 June 2026 2 Days
Understanding the business problem	Cause effect analysis Cost of each breakdown Understanding Types Of Breakdowns	29 June 2026 2 days
Get the Data	Download and import Row data	30 June 2026 1 day
Explore and pre-process data	Import the data Remove duplicate, Check Null Value Check Data types Check Skewness Drop Outliers by using IQR Method Check Corelations normalize numeric variance,	1 July 2026 4 days
Choosing the python platform	Jupyter Notebook / VS Code	1 July 2026
Create Features	'Air temperature [K]', 'Process temperature [K]', 'Rotational speed [rpm]', 'Torque [Nm]', 'Tool wear [min]',	1 July 2026

	'Type_L', 'Type_M'	
EDA	Generate Pearson correlation matrices and component feature plots.	5 July 2026 2 day
Create Model	Create Model And Test Performance RandomForestClassifier DecisionTreeClassifier LogisticRegression XGBClassifier	6 July 2026 3 day
Model Evaluation	Run confusion matrices, Classification report, F1 Score, Train Test accuracy Overfitting Status	9 July 2026 2 days
Report Writing	Document hyperparameter selections, operational findings	11 July 2026 2 days
Project submission	Bundle script notebooks, metrics report, and final documentation.	14 July 2026
